



SUMMER INTERNSHIP PRESENTATION

SmartSplit

Expense Management & Payment Settlement Platform

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Organization: BitePay

Duration: 7 June 2026 – 7 August 2026

Domain: AI, Data Analytics & Software Development

Internship Overview



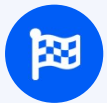
Organization

BitePay — Tech that Serves



Role

AI, Data Analytics & Software Dev. Intern



Duration

2 Months (7 Jun – 7 Aug 2026)



Domain

Fintech — Payments & Data Analytics

Key Areas of Exposure

- Full-Stack Web Application Development
- REST API Design (Node.js + Express.js)
- Database Design with MongoDB & Mongoose
- JWT Authentication & Secure Password Handling
- Payment Gateway Integration (Razorpay)
- Real-World Fintech Product Workflow

About BitePay



"Tech that Serves"

MSME Registered · 2025



BitePay is a fintech company building digital payment and financial-utility products.



Works across AI/Data Analytics and software engineering to power its payment-related software.



Focuses on practical, real-world product building — interns contribute directly to live features.

My Role: As an intern in AI, Data Analytics & Software Development, I contributed to building a fintech-style expense and payment-settlement product from the ground up.

Objectives of the Internship



Build a complete full-stack application end-to-end, not isolated components



Gain hands-on experience designing and consuming REST APIs



Understand secure authentication using JWT and password hashing



Learn schema design and data modeling with MongoDB



Integrate a real third-party payment gateway (Razorpay) into a live workflow



Develop practical exposure to a professional fintech product environment

Project: SmartSplit

A full-stack expense-splitting and payment-settlement application

What it does

SmartSplit simplifies managing shared expenses. Users create shared expenses, split amounts equally or by custom shares among participants, track balances on a dashboard, and settle outstanding amounts online.

Payments are handled through Razorpay Checkout, and every payment is verified on the backend rather than trusted from the frontend alone — closing a common security gap in payment flows.



User Accounts

Registration & secure login



Expense Splitting

Equal or custom shares among participants



Dashboard

Track expenses & outstanding balances



Payments

Settle shares via Razorpay Checkout

Technologies & Tools Used



Frontend

React.js, JavaScript, HTML, CSS



Backend

Node.js, Express.js (REST APIs)



Database

MongoDB with Mongoose ODM



Auth & Security

JWT authentication, bcrypt password hashing



Payments

Razorpay Payment Gateway & Checkout

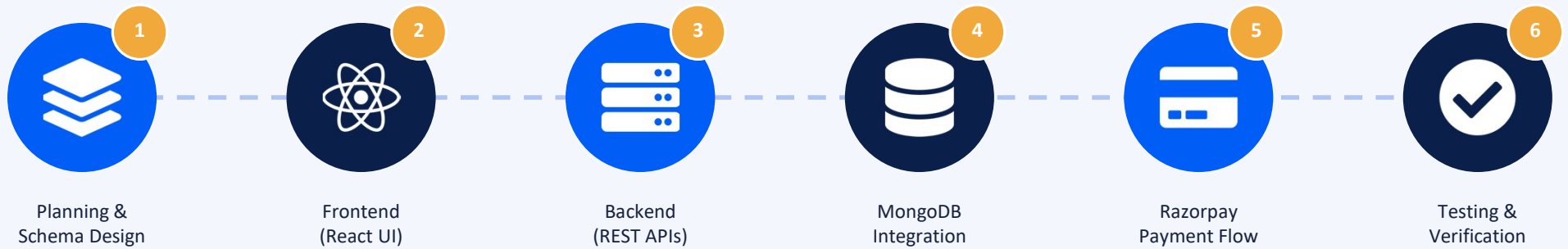


Dev Tools

VS Code, npm, Nodemon, Git/GitHub, dotenv, CORS

Development Workflow

How the frontend, backend, database and payment workflow came together



Integrated the frontend, backend APIs, database and payment workflow into a single working application — rather than building components in isolation.

Challenges & Key Learnings



Challenges

- MongoDB Atlas DNS/SRV resolution and TLS connection issues when connecting the Node.js backend to the Atlas cluster
- Understanding how connection strings, DNS resolution, IP whitelisting and Node.js config affect a live backend
- Integrating the Razorpay flow — creating orders, launching Checkout from React, and securely verifying payments on the backend



Key Learnings

- How to build a full-stack app end-to-end — connecting React, Express, and MongoDB instead of isolated pieces
- Practical experience designing REST APIs and implementing JWT-based authentication
- Designing MongoDB schemas for a real application
- How to integrate and secure a real online payment workflow

Internship Outcomes

After completing the two-month internship, I was able to:



Deliver a working full-stack SmartSplit prototype — React dashboard + Node/Express backend + MongoDB



Enable users to register, log in, and manage shared expenses end to end



Support equal and custom expense splitting among participants



Let users view outstanding balances and initiate real payments via Razorpay



Implement backend payment verification, not just frontend trust



Gain practical, integrated experience across auth, database design and fintech payment workflows



Conclusion

Successfully completed a 2-month Summer Internship in AI, Data Analytics & Software Development at BitePay, from 7 June 2026 to 7 August 2026.

Built SmartSplit, a full-stack expense-splitting and payment-settlement application connecting React, Express, MongoDB and Razorpay — gaining hands-on experience in authentication, database design, REST APIs and real-world fintech payment integration.

“This internship strengthened my ability to connect theoretical knowledge with real-world, end-to-end product development — and deepened my interest in building practical, full-stack fintech solutions.”